How Arm Angle Influences Pitcher Performance in Major League Baseball

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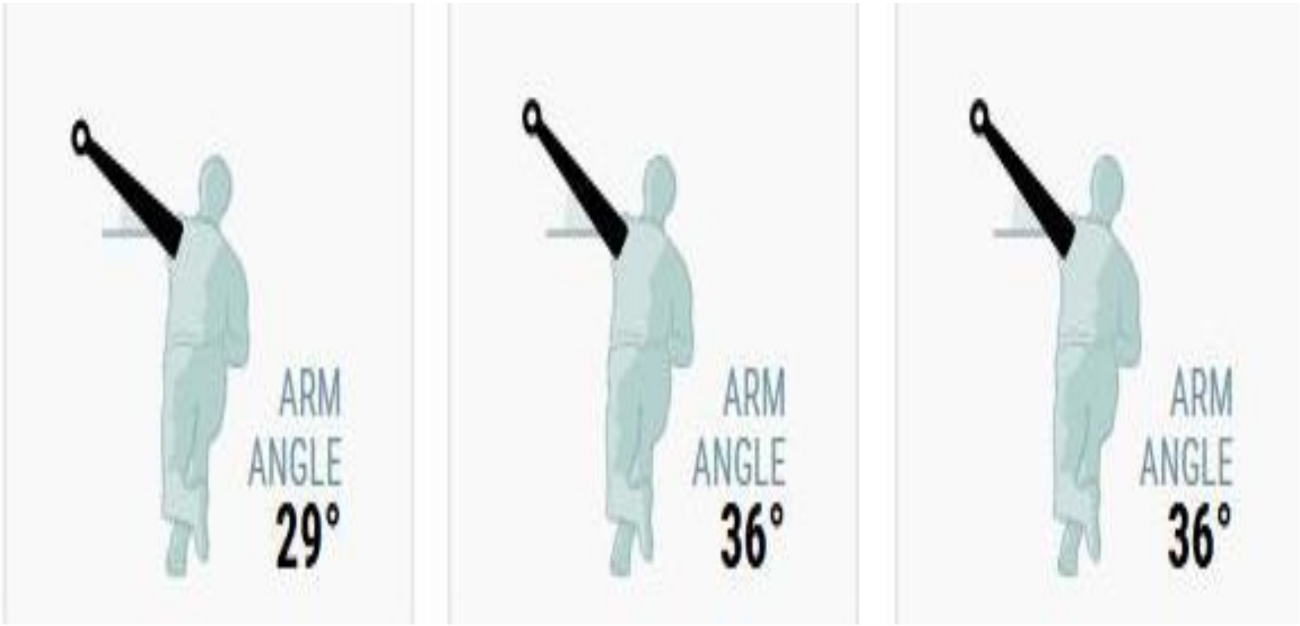
What is a Pitcher?

- Starts by throwing the ball at the hitter
- Prevent hitter by missing their swing, strikeouts, or weak hit
- Very complex mechanics to pitch well

The Research motivation

- Pitchers' performance is very important
- Lot of research on factors impacting performance
  - The front leg is important to a pitch <- Brewster (2019)
- Under researched



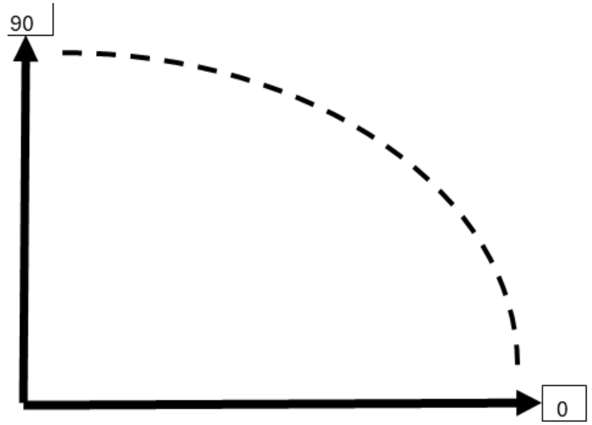
What is the Data?

- Databank: **BaseballSavant**
- Sample size: **305**
- Data-type: **Pitch statistics per Pitcher**
- Years: **2024,2023,2022** <- 3 seasons

What are the Variables used?

- **Dependent Variable:** Whiff Rate
- **Main Independent Variable:** Arm Angle
- **Control Variables:**
  - Type of pitches (Fastballs, Breaking, Off-speed)
  - Descriptive pitch variables (avg speed, avg spin)
  - Number of Pitches
  - Pitching Hand
  - Season dummy variables

Intercept
pitch_count
arm_angle
fastball_avg_speed
fastball_avg_spin
breaking_avg_speed
breaking_avg_spin
offspeed_avg_speed
offspeed_avg_spin
Hand
FB/100
Breaking/100
Offspeed/100
2024
2023



Regression 1

Metric	Value
Residual DF	303
R2	0.0001586
Adjusted R2	-0.0031412
Std. Error Es	0.0397458
RSS	0.4786586

Predictor	Estimate	Confidence Interval: Lower	Confidence Interval: Upper	Standard Error	T-Statistic	P-Value
Intercept	0.2512405	0.232783432	0.269697538	0.009379422	26.7863497	6.91E-82
arm_angle	-5.001E-05	-0.000498925	0.00039891	0.000228129	-0.21920598	0.826637

- Arm angle = **insignificant**
- Not a determinant of the Whiff Rate
- R-squared: **0.0001586**

Final Regression

- R-squared: **0.459**
- Arm angle is still **insignificant**
- Significant Variables:
  - **Pitch Count**
  - **Fastball Average Speed and Spin**
  - **Hand (Right = 1)**
- Other variables help to control for **omitted-variable bias**

Predictor	Estimate	Standard Error	T-Statistic	P-Value
Intercept	2.2108886	3.45128645	0.64059839	0.522289
pitch_count	1.556E-05	4.94557E-06	3.1465481	0.001824
arm_angle	-0.00015	0.000186779	-0.80335148	0.42243
fastball_avg_speed	0.0085592	0.001137846	7.52225896	6.79E-13
fastball_avg_spin	0.0001052	1.4738E-05	7.13933896	7.57E-12
breaking_avg_speed	-0.0002555	0.00078295	-0.32631312	0.744423
breaking_avg_spin	-1.717E-05	9.17667E-06	-1.87123573	0.062319
offspeed_avg_speed	-0.0011558	0.000756788	-1.52722964	0.127794
offspeed_avg_spin	3.62E-06	6.3956E-06	0.56607182	0.571783
Hand	-0.0129774	0.004104797	-3.16153251	0.001736
FB/100	-2.887296	3.44377041	-0.83841129	0.40249
Breaking/100	-2.8373765	3.444776022	-0.82367518	0.410801
Offspeed/100	-2.7991274	3.445314141	-0.81244476	0.417203
2024	-0.0063637	0.004220736	-1.5077123	0.132717
2023	-0.0010157	0.004245769	-0.23922023	0.811104

Metric	Value
Residual DF	290
R2	0.4592606
Adjusted R2	0.4331559
Std. Error Estimate	0.0298774
RSS	0.2588707

Variable Statistics

	player_age	pitch_count	arm_angle	fastball_avg_speed	fastball_avg_spin	breaking_avg_speed	breaking_avg_spin
Average:	29.02	2588.68	39.89	92.84	2256.15	81.99	2454.17
STD.Dev	3.94	359.79	9.99	2.30	150.62	3.28	226.41
Min	22	1825	10.9	86.8	1885	71.2	1941
Max	43	3281	70.1	98.9	2704	89.9	3088

	offspeed_avg_speed	offspeed_avg_spin	Hand	WHIFF %	FB/100	Breaking/100	Offspeed/100	2024	2023
Average:	85.71	1690.44	73.44%	24.92%	56.28%	29.11%	14.62%	33.77%	32.46%
STD.Dev	2.86	290.01	44.24%	3.97%	10.00%	11.50%	9.67%	47.37%	46.90%
Min	69.7	640	0	0.164	0.337	0.01	0	0	0
Max	92.9	2477	1	0.386	0.863	0.609	0.425	1	1

What does all this mean?

- Arm Angle is an **insignificant measure on the Whiff Rate**
- Factors that matter:
  - **Number of pitches**
  - **Fastball Average Speed and Spin**
  - **Being Left-Handed**
- Only Fastball descriptive variables mattered
- Knee extension and forward trunk tilt -> **positively correlated with velocity**

References

- https://baseballsavant.mlb.com/leaderboard/custom?year=2024%2C2023%2C2022&type=pitcher&filter=&min=q&selections=player_age%2Cpitch_count%2Cwhiff_percent%2Cpitch_hand%2Carm_angle%2Cn_ff_formatted%2Cff_avg_speed%2Cff_avg_spin%2Cn_fastball_formatted%2Cfastball_avg_speed%2Cfastball_avg_spin&chart=false&x=arm_angle&y=arm_angle&r=no&chartType=beeswarm&sort=pitch_count&sortDir=desc
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Metric	Value
Residual DF	293
R2	0.4321474
Adjusted R2	0.4108287
Std. Error Estimate	0.0304601
RSS	0.2718507

Predictor	Estimate	Standard Error	T-Statistic	P-Value
Intercept	-0.6642652	0.079021235	-8.40616099	1.87E-15
pitch_count	1.536E-05	5.03435E-06	3.05201251	0.002481
arm_angle	-0.0001558	0.000190363	-0.81830394	0.413849
fastball_avg_speed	0.0092022	0.001145589	8.0327096	2.35E-14
fastball_avg_spin	8.436E-05	1.38708E-05	6.08180343	3.7E-09
breaking_avg_speed	0.0001139	0.000774137	0.14718475	0.883087
breaking_avg_spin	-1.443E-05	9.10014E-06	-1.58598548	0.113821
offspeed_avg_speed	-0.0015755	0.000760381	-2.07201119	0.039139
offspeed_avg_spin	5.219E-06	6.46279E-06	0.80759984	0.419976
Hand	-0.0137367	0.004171798	-3.29275141	0.001114
2024	-0.0060032	0.004279035	-1.40292996	0.161696
2023	-0.0007998	0.004321235	-0.1850768	0.853297